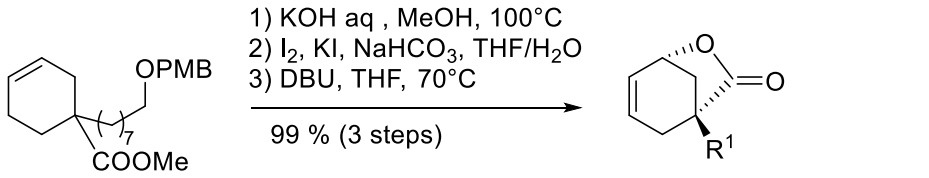
**Cume for Organic Chemistry, September 25, 2020**
**75/100 points to pass**

**1) Give the full names and structures of the following abbreviations (3 point each)**

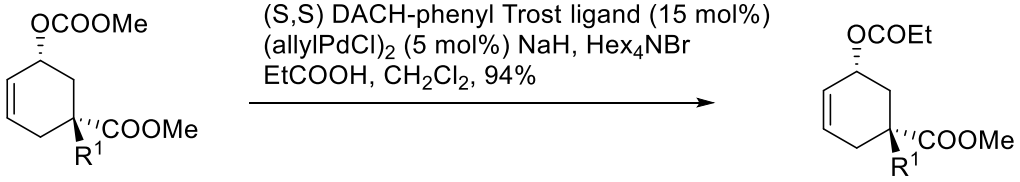
- **LDA**
- **HMPA**
- **PMB**
- **DBU**
- **(S,S)-DACH-phenyl TROST ligand**
- **DMAP**
- **DIBAL**
- **TBS**
- **$Hex_4NBr$**
- **TBSCl**

**2) Give a mechanism for the following transformations (25 points)**




$R^1$ = $-(CH_2)_8$-OPMB

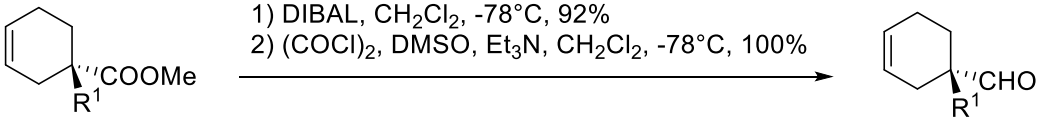
**3) Explain how the Pd-mediated resolution of the racemic carbonate works (25 points)**




$R^1$ = $-(CH_2)_8$-OPMB

98% ee

**4) Give the mechanism of the second part of the reaction including the formation of the active ingredients (20 points) – this is a name reaction, do you know how it's called ? (10 bonus points)**




$R^1$ = $-(CH_2)_8$-OPMB